# ChangHo Choi

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## Education

**Korea Advanced Institute of Science and Technology** — Daejeon, South Korea
Ph.D. in Electrical Engineering — *Mar. 2026 – Present*
Statistical Inference & Information Theory Lab, Advisor: Prof. Junmo Kim

**Carnegie Mellon University** — Pittsburgh, PA, USA
Visiting Scholar, Software and Societal Systems Department — *Aug. 2026 – Present*

**Korea Advanced Institute of Science and Technology** — Daejeon, South Korea
M.S. in Electrical Engineering — *Mar. 2024 – Feb. 2026*
Statistical Inference & Information Theory Lab, Advisor: Prof. Junmo Kim

**Korea University** — Seoul, South Korea
B.S. in Cyber Defense, GPA: 4.07/4.5 — *Mar. 2020 – Feb. 2024*

## Research & Industry Experience

**PYLER** — Remote Intern — *Oct. 2026 – Present*

- Built a multimodal video retrieval pipeline combining embedding-based search, multimodal LLM verification, and web-search agents to return timestamped moments from natural-language queries.

**KRAFTON** — AI Foundation Model Team
Machine Learning Engineer, Data Team Intern — *Jan. 2026 – Aug. 2026*

- Contributed to **Raon-1-mini** (30B-A4B MoE LLM) and **Raon-Speech-9B** (9B Speech LLM) **trained from scratch**, developing large-scale data pipelines.
- **Speech LLM:** Built and curated large-scale speech–text datasets for Raon-Speech and its full-duplex extension through ASR/TTS pipelines, alignment, dialogue processing, and quality filtering. Scaled processing across a large B300 GPU cluster using vLLM and Slurm, and evaluated model performance and training data quality using proxy models.
- **Text LLM:** Built large-scale pretraining corpora for Raon-1-mini through synthesizing, web crawling, curation, deduplication, preprocessing, and quality filtering.

**Machine Learning & Vision Lab, Korea University** — Seoul, South Korea
Undergraduate Student Researcher, Advisor: Prof. Hyunwoo J. Kim — *Dec. 2022 – Sep. 2023*

**Computer Systems Security Lab, Korea University** — Seoul, South Korea
Undergraduate Student Researcher, Advisor: Prof. Youngjoo Shin — *Dec. 2021 – Mar. 2022*

## Selected Publications

**Continuous Audio Thinking for Large Audio Language Models** — *NeurIPS 2026*
Gyojin Han*, Dong-Jae Lee*, **Changho Choi***, Jongsuk Kim*, Junmo Kim
– Audio-expert distillation enables continuous latent thinking without additional autoregressive decoding.

**Raon-Speech Technical Report** — *arXiv*
KRAFTON
– A 9B English–Korean SpeechLM for speech understanding and generation, extended to full-duplex conversation.

**B4DL: A Benchmark for 4D LiDAR LLM in Spatio-Temporal Understanding** — *ACM MM 2025*
**Changho Choi***, Youngwoo Shin*, Gyojin Han, Dong-jae Lee, Junmo Kim
– A benchmark, data pipeline, and MLLM for spatio-temporal reasoning directly from raw 4D LiDAR.

**Iterative Optimization of SDS Loss for Video Generation from Multi-Object Sketches** — *NeurIPS Workshop 2024*
Donghun Kim, **Changho Choi**, Junmo Kim
– Iterative object-wise and joint SDS optimization animates multi-object sketches using video diffusion priors.
* Equal contribution.

## Technical Skills

**Programming:** Python, C/C++
**Deep Learning & Systems:** PyTorch, Hugging Face Transformers/Datasets, vLLM, Slurm, Linux
**Speech & Audio:** ASR/TTS pipelines, speaker diarization, forced alignment, full-duplex speech data processing
**Languages:** Korean, English

## Teaching Experience

**Teaching Assistant, KAIST**

Signals and Systems (EE202) *Spring & Fall 2025*
Programming Structures for Electrical Engineering (EE209) *Fall 2024*

## Scholarships

**Korea University**

Full Tuition Scholarship for Academic Excellence *2020 – 2023*
Admitted with the highest entrance score to Korea University.